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README FILE

Particle in a Box Simulator:

This project simulates the quantum mechanical model known as the **particle in a box**, a fundamental concept in quantum mechanics. It visualizes wavefunctions and corresponding energy levels for a particle confined in a one-dimensional infinite potential well.

Overview:

The simulation computes and plots:

- Wavefunctions ($Wf(n,a)$)
- Energy levels ($Energy(n)$)

for different quantum numbers (n).

This model is a classic example of quantization in the particle in a box system.

Features:

1. Computes normalized wavefunctions
2. Calculates discrete energy levels
3. Plots multiple quantum states ($n = 1$ to 5)
4. Displays energy values directly in the legend

Formulas Used:

Wavefunction –

$$Wf\ n(a) = ((2/L)**(1/2))\sin(n*\pi*x/L)$$

Energy level-

$$\text{Energy}(n) = \frac{(nh)^2}{8mL^2}$$

Where:

- (n): Quantum number (1, 2, 3, ...)
- (h): Planck's constant
- (m): Mass of the particle (electron)
- (L): Length of the box

Requirements:

Install the required Python libraries:

```
pip install numpy matplotlib
```

How to Run:

1. Clone or download this repository
2. Run the Python script:

```
python particle_in_box.py
```

3. A plot will be displayed showing wavefunctions for different energy states.

The graph shows:

- Wavefunctions for ($n = 1$) to ($n = 5$)
- Nodes increasing with quantum number
- Energy values in Joules displayed in the legend

Key Insights:

- Energy is **quantized** and increases with (n^2)
- Higher (n) $\rightarrow$ more oscillations in the wavefunction
- The particle cannot exist outside the box
- Wavefunctions are zero at the boundaries

References:

- Introductory Quantum Mechanics textbooks

- Standard solutions of the Schrödinger equation